

Apply spring reasoning

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people? Copy or complete the first step.

2. Order 0.72, $\frac{7}{10}$ and 73%.

3. Calculate $\frac{5}{6} + \frac{7}{12}$.

4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people? Copy or complete the first step.
Answer $1\frac{3}{4}$ kg.
2. Order 0.72, $\frac{7}{10}$ and 73%.
Answer $\frac{7}{10}$, 0.72, 73%.
3. Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer $1\frac{5}{12}$.
4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.
Answer Name the domain first, represent the relationship, then choose and check the calculation.

Apply spring reasoning

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people?

6. Check this result using an inverse, estimate or known fact: Calculate $\frac{5}{6} + \frac{7}{12}$.

2. Order 0.72, $\frac{7}{10}$ and 73%.

7. Prove or explain your reasoning: Find 37.5% of 640.

3. Calculate $\frac{5}{6} + \frac{7}{12}$.

8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.

4. Solve this independently and show every stage: A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Order 0.72, $\frac{7}{10}$ and 73%.

10. Write and solve a similar question to: Calculate $\frac{5}{6} + \frac{7}{12}$.

Teacher answers and checking prompts

1. A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people?
Answer $1\frac{3}{4}$ kg.
2. Order 0.72, $\frac{7}{10}$ and 73%.
Answer $\frac{7}{10}$, 0.72, 73%.
3. Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer $1\frac{5}{12}$.
4. Solve this independently and show every stage: A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people?
Answer $1\frac{3}{4}$ kg.
5. Use a second method or representation for: Order 0.72, $\frac{7}{10}$ and 73%.
Answer Expected result: $\frac{7}{10}$, 0.72, 73%. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer Expected result: $1\frac{5}{12}$. A valid check must be shown.
7. Prove or explain your reasoning: Find 37.5% of 640.
Answer 240. The explanation must show why the method works.
8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.
Answer Name the domain first, represent the relationship, then choose and check the calculation.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Apply spring reasoning

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find 37.5% of 640.

6. Create a new example that tests the same idea as: Calculate $\frac{5}{6} + \frac{7}{12}$.

2. Prove the answer to this question, rather than only calculating it: A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Order 0.72, $\frac{7}{10}$ and 73%.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $1\frac{5}{12}$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find 37.5% of 640.
Answer 240. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A recipe uses $\frac{3}{4}$ kg for 6 people. How much is needed for 14 people?
Answer Expected result: $1\frac{3}{4}$ kg. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Order 0.72, $\frac{7}{10}$ and 73%.
Answer Expected result: $\frac{7}{10}$, 0.72, 73%. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $1\frac{5}{12}$.
Answer One valid reconstruction is: Calculate $\frac{5}{6} + \frac{7}{12}$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.
Answer Name the domain first, represent the relationship, then choose and check the calculation.
6. Create a new example that tests the same idea as: Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.